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Microchiropterans have a diminutive cerebral cortex, not an enlarged cerebellum, compared to megachiropterans and other mammals

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Abstract

Small echolocating bats are set apart from most other mammals by their relatively large cerebellum, a feature that has been associated to echolocation, as it is presumed to indicate a relatively enlarged number of neurons in the cerebellum in comparison to other brain structures. Here we quantify the neuronal composition of the cerebral cortex, cerebellum and remaining brain structures of seven species of large Pteropodid bats (formerly classified as megachiropterans), one of which echolocates, and six species of small bats (formerly classified as microchiropterans), all of which echolocate. This chiropteran data is compared to 60 mammalian species in our dataset to determine whether the relatively large cerebellum of the small echolocating bats, and possibly that of the echolocating Pteropodid, contains a relatively enlarged number of neurons. We find no evidence that the distribution of neurons differs between microchiropterans and megachiropterans, but our data indicate that microchiropterans, like the smallest shrew in our dataset, have diminutive cerebral cortices, which makes the cerebellum appear relatively large. We propose that, in agreement with the diminutive brain size of the earliest fossil mammals, this is a plesiomorphic, not a derived, feature of microchiropteran brains. The results of this study also reveal important neural characteristics related to the phylogenetic affinities and relationships of the chiropterans.

KEYWORDS
brain size, chiroptera, echolocation, evolution, numbers of neurons, RRID: AB_11204707, RRID: AB_2307445

1 | INTRODUCTION

Bats (order Chiroptera) are the only mammals capable of true, sustained (flapping) flight, which requires a high-energy expenditure (Thomas, 1975; Thomas & Suthers, 1972) that may affect the energetic budget of the brain. Bats were initially considered a monophyletic group subdivided into the larger-bodied, predominantly fruit-eating, nonecholocating megachiropteran suborder, and the smaller-bodied, insectivorous, echolocating microchiropteran suborder (Smith & Madkour, 1980); however, the phylogeny of the Chiroptera is a subject of contention. Resolving bat phylogeny has a practical implication on our understanding of the evolution of echolocation, that is, the capability of navigation using an acoustic image of the environment created from one's own emitted sounds, which has long been considered a

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derived evolutionary feature (Hartridge, 1945; Thiagavel et al., 2018) believed to involve a series of structural and functional changes of the nervous system in the Chiroptera (Safi & Dechmann, 2005; Simmons, Fenton, & O'Farrell, 1979; Ulanovsky & Moss, 2008).

One view based initially on divergent neural and genital traits (Pettigrew, 1986; Pettigrew et al., 1989) proposes that the Chiroptera are diphyletic, the megachiropterans and microchiropterans having evolved from unrelated ancestors at different times. This view aligns the megachiropterans as a recent evolutionary branch of the Dermoptera (the acknowledged closest mammalian relatives of the Primates), and aligns the microchiropterans with the eulipotyphlan shrews (Dell et al., 2010). The other view proposes that the Chiroptera is a natural monophyletic group, based on molecular and morphological characters (Jones, Purvis, Maclarnon, Bininda-emonds, & Simmons, 2002; Teeling et al., 2005). The monophyletic proposal (O'Leary et al., 2013) subdivides the chiropterans into two main groups: (a) the Yangochiroptera (which comprise the majority of small echolocating, previously "microchiropteran" families) and (b) the Yinpterochiroptera (or the Pteropodiformes), which is comprised of the only megachiropteran family, Pteropodidae, plus the Rhinopomatidae, Rhinolophidae, Hipposideridae, and Megadermatidae families previously classified as microchiropteran (Agnarsson, Zambrna-Torrelio, Flores-Saldana, & May-Collado, 2011; Hawkins et al., 2019; Lei & Dong, 2016; Tsagkogeorga, Parker, Stupka, Cotton, & Rossiter, 2013). Importantly, this monophyletic proposal is at odds with the most obvious morphological and behavioral characteristics of small and echolocating versus large and nonecholocating bats, and does not provide clarity on the relationship of the order Chiroptera to other mammalian species.

Irrespective of their phylogenetic relationships, small echolocating bats (including those assigned to the Yinpterochiroptera) have a distinguishing brain characteristic: their cerebellum is relatively much larger than the cerebellum of most other mammalian species, comprising approximately 25% of total brain mass rather than the typical 10–15% (Clark, Mitra, & Wang, 2001; Maseko, Spocter, Haagensen, & Manger, 2012). This apparent enlargement of the cerebellum has been proposed to be related to the neural mechanisms required for the production and interpretation of the vocalizations used for echolocation (Maseko et al., 2012), possibly akin to the relatively larger cerebellum of two other groups: cetaceans (who also echolocate; Paulin, 1993; Marino, Rilling, Lin, & Ridgwway, 2000; Ridgway, 2000), and elephants (who are not known to echolocate, but do have a form of specialized infrasonic vocalizations; Herbst et al., 2012).

The absolute, relative and residual volumes and surface areas of neural structures were once the mainstay of comparative neural studies, and many inferences about neuronal functional adaptations were made based on the premise that enlarged structures were an indication of associated advantageous functional enhancements that had been positively selected for (Jerison, 1973). Thus, the relative enlargement of the cerebral cortex as larger mammalian brains evolved was considered as evidence of a functional cortical takeover (Clark et al., 2001), while the cerebellum lagged behind—with the exception of microchiropterans, cetaceans and proboscideans, who still share relatively large cerebella despite substantial differences in absolute brain mass (Maseko et al., 2012).

With the development of the isotropic fractionator, a non-stereological, rapid method of counting cells in any dissectible organ (Herculano-Houzel & Lent, 2005), we have undertaken a systematic investigation of the cellular composition of the brains of dozens of mammalian species, belonging to a wide range of clades (marsupials, afrotherians, glires, scandentians, primates, carnivorans, eulipotyphlans, and artiodactyls; Azevedo et al., 2009; Dos Santos et al., 2017; Gabi, Collins, Wong, Kaas, & Herculano-Houzel, 2010; Herculano-Houzel, Collins, Wong, & Kaas, 2007; Herculano-Houzel et al., 2011; Herculano-Houzel et al., 2014; Herculano-Houzel, Catania, et al., 2015; Herculano-Houzel, Kaas, et al., 2015; Herculano-Houzel, Mota, & Lent, 2006; Jardim-Messeder et al., 2017; Kazu, Maldonado, Mota, Manger, & Herculano-Houzel, 2014; Neves Jr et al., 2014; Sarko, Catania, Leitch, Kaas, & Herculano-Houzel, 2009). These quantitative analyses demonstrate that the absolute and relative mass or volume of brain structures cannot always be used as a proxy for absolute and relative numbers of neurons in those structures. For example, while mammalian species with larger brains do tend to have relatively larger cerebral cortices than smaller-brained species, all species retain a relative distribution of 10–25% of all neurons in the cerebral cortex, and 75–85% of neurons are found in the cerebellum—regardless of the absolute or relative mass of these structures (Herculano-Houzel, 2010). The only known outliers to date are the African elephant (*Loxodonta africana*; Herculano-Houzel et al., 2014) and the brown bear (*Ursus arctos*; Jardim-Messeder et al., 2017), with 98 and 96% of brain neurons found in the cerebellum, but this appears to have evolved through different mechanisms. The African elephant has the expected number of cortical neurons for both its brain mass and its number of neurons in the brainstem, diencephalon and striatum, and an unusually high number of neurons in the cerebellum for its mass, meaning that it has a truly enlarged cerebellum (Herculano-Houzel et al., 2014). In contrast, the brown bear has the expected number of neurons in its cerebellum for its mass and for its number of neurons in the brainstem, diencephalon and striatum, but has an extremely low number of neurons in the cerebral cortex, despite its proportional mass, which we speculate to be related to the same energetic insufficiency that leads this species to hibernate (Jardim-Messeder et al., 2017). Thus, just like a relatively larger cerebral cortex is not necessarily an indication of a cerebral cortex with proportionately more neurons (Herculano-Houzel, 2010), a relatively large cerebellum is not necessarily an indication of a potentially functionally enlarged and enhanced cerebellum—that is, a cerebellum that has gained an unusually large number of neurons.

Here we address whether or not the relatively large cerebellum of small echolocating bats, including Rhinolophoidea (Maseko et al., 2012), is truly an enlarged cerebellum, that is, whether it contains an overly large number of neurons for its mass, as well as relative to the number of neurons in the cerebral cortex. Such an enlargement, if confirmed, would favor suggestions that the relatively large cerebellum of these animals has been selected for reasons possibly relating to echolocation. On the other hand, finding that the relatively large cerebellum of echolocating small bats is not an enlarged cerebellum in its cellular composition would falsify that notion. Because we are interested in comparing numbers of neurons and their ratios across brain structures in select species, and aim to do so exactly in the context of a contested phylogeny, we limit our analysis to the raw data without applying any methods for "correcting" for relatedness in the dataset, as will be explored in detail in the Discussion.

2 | MATERIALS AND METHODS

Here we analyze the brain cellular composition of 13 species of chiropterans, comprising six species once classified as microchiropterans: three Rhinolophoidae and three Yangochiroptera, and seven species of Pteropodidae, once classified as megachiropterans (Table 1). The latter include *Rousettus aegyptiacus*, a species that uses clicks of the tongue for echolocation, in contrast to the laryngeal echolocation used by microchiropterans (e.g., Jones & Teeling, 2006). We use the isotropic fractionator (Herculano-Houzel, Kaas, et al., 2015; Herculano-Houzel & Lent, 2005) to determine the numbers of neuronal and non-neuronal cells that constitute each of four main brain parts: cerebral cortex, cerebellum, olfactory bulb (where available) and the rest of the brain (comprising hindbrain, midbrain, diencephalon and striatum). We then use the data to test whether the relatively large cerebellum of small echolocating bats contains an enlarged proportion of all brain neurons.

2.1 | Animals

All animals were captured from wild populations and all were adults, showing no signs of advanced age, and appeared neurotypical. The animals were caught in Kenya, South Africa, and the Democratic Republic of the Congo, and were treated and used according to the guidelines of the University of the Witwatersrand Animal Ethics Committee (clearance number 2008/36/1). Permissions to capture and euthanize these animals were obtained from the Kenya National Museums and the Kenyan Wildlife Services (*Cardioderma cor*, *Coleura afra*, *Triaenops persicus*, *Hipposideros commersoni*, *Chaerephon pumilus*, *Epomophorus wahlbergii*), the Gauteng Nature Conservation Directorate (*Miniopterus schreibersii*), the Limpopo Provincial Nature Conservation Directorate (*Rousettus aegyptiacus*), and the University of Kisangani (*Megaloglossus woermanni*, *Hypsignathus monstrosus*, *Epomops franqueti*, *Casinycteris argynnis*, *Scotonycteris zenkeri*). For all specimens except *Chaerophon pumilus*, we had only one hemisphere available for quantitative analysis, with the other hemisphere used for histological examination. For most animals, then, data are necessarily expressed as twice the numbers obtained for the available hemisphere, to make them comparable to other mammalian species in our dataset; for *C. pumilus*, numbers provided refer to the sum of both hemispheres.

2.2 | Dissection

All animals were anesthetized (overdose of sodium pentobarbital, 100 mg/kg) within 1 hr of capture and perfused transcardially with 0.9% normal saline followed by 4% paraformaldehyde in 0.1M phosphate buffer (PB). Brains were removed from the skull after transecting the spinal cord at the level of the foramen magnum, weighed, postfixed in 4% paraformaldehyde in 0.1M PB overnight,

TABLE 1 Dataset

Species	Micro/mega	Family	Clade	*n*	M_{BODY}, g	M_{BRAIN}, g	N_{BRAIN}
Cardioderma cor	Microbat	Megadermatidae	Rhinolophoidea	2 (1H each)	26.0	0.562	81,331,192
Coleura afra	Microbat	Emballonuridae	Yangochiroptera	2 (1H each)	11.5	0.244	58,580,214
Triaenops persicus	Microbat	Hipposideridae	Rhinolophoidea	2 (1H each)	13.7	0.251	69,762,924
Hipposideros commersoni	Microbat	Hipposideridae	Rhinolophoidea	2 females (1H each)	101.9	0.588	66,718,833
Chaerephon pumilus	Microbat	Molossidae	Yangochiroptera	2 (1 female) (2H each)	5.4	0.208	34,958,391
Miniopterus schreibersii	Microbat	Miniopteridae	Yangochiroptera	2 (1H each)	11.6	0.219	55,658,777
Epomophorus wahlbergi	Megabat	Pteropodidae	Pteropodidae	2 females (1H each)	74.0	1.224	161,754,633
Rousettus aegyptiacus	Megabat	Pteropodidae	Pteropodidae	2 (1 female) (1H each)	120.0	1.778	172,232,047
Megaloglossus woermanni	Megabat	Pteropodidae	Pteropodidae	2 (1H each)	17.9	0.550	65,728,366
Hypsignathus mostrosus	Megabat	Pteropodidae	Pteropodidae	2 (1 female) (1H each)	310.0	3.459	274,519,253
Epomops franqueti	Megabat	Pteropodidae	Pteropodidae	2 (1H each)	120.0	2.044	186,002,499
Casinycteris argynnis	Megabat	Pteropodidae	Pteropodidae	2 (1H each)	27.0	0.892	99,016,077
Scotonycteris zenkeri	Megabat	Pteropodidae	Pteropodidae	2 (1H each)	16.1	0.588	65,594,971

Note: "Micro/mega" refers to the traditional classification into echolocating (micro) versus nonecholocating, or laryngeal echolocating (mega) bats. For the purposes of this study, bat families are grouped into the three consensus clades, Rhinolophoidea, Pteropodidae, and Yangochiroptera, without specifying whether Rhinolophoidea should cluster with Pteropodidae as Yinpterochiroptera (Tsagkogeorga et al., 2013) or with Yangochiroptera as a sister group to Pteropodidae (Argnarsson et al., 2011).

cryoprotected in 30% sucrose in 0.1M PB, and stored in an anti-freeze solution at −20°C until use.

The cerebellum was dissected by cutting the cerebellar peduncles at the surface of the brainstem. Cerebral cortex in all animals was defined as all cortical regions lateral to the olfactory tract, including hippocampus and piriform cortex, and dissected from each hemisphere by peeling it away from the subcortical structures, as described earlier (Herculano-Houzel et al., 2006). In this manner, the cerebral cortex includes the underlying white matter down to, but not including, the surface of the striatum, therefore excluding the internal capsule. In some cases, the hippocampus was dissected and weighed separately, but the numbers reported here for "cerebral cortex" always include the hippocampus. The olfactory bulb was dissected and processed separately when possible. All other brain structures (the ensemble of brainstem, diencephalon, and striatum) were pooled and processed together as "rest of brain," consistent with our practice with all other mammalian species analyzed to date.

2.3 | Isotropic fractionator

Total numbers of cells, neurons, and nonneuronal cells were estimated as described previously using the isotropic fractionator method (Herculano-Houzel & Lent, 2005). Briefly, each dissected brain structure was weighed and dissolved in a saline detergent solution in a glass homogenizer, which turns the tissue into an isotropic suspension of isolated nuclei of known, defined volume, kept homogeneous by agitation, and stained with the fluorescent DNA marker DAPI (4′,6-diamidino-2-phenylindole dihydrochloride; RRID: AB_2307445). The total number of nuclei in the suspension—and therefore the total number of cells in the original tissue—is estimated by determining the density of nuclei in typically four 10 μl aliquots under a fluorescence microscope using a 40× objective. Values of total number of cells are calculated as the product of the average nuclei/ml of the aliquots multiplied by the total volume of the suspension. This consistently yields a coefficient of variation of 0.10, and never more than 0.15, across samples for the same structure. Once the total number of cells in a structure is known, the proportion of neurons is determined by immunocytochemical detection of neuronal nuclear antigen (NeuN), expressed in all nuclei of most neuronal cell types and not in nonneuronal cells (Mullen, Buck, & Smith, 1992), using Cy3-conjugated rabbit polyclonal anti-NeuN antibody (RRID: AB_11204707). Numbers of nonneuronal cells are derived by subtraction. For consistency, reported total brain mass and numbers of brain neurons exclude the olfactory bulbs, following our practice since 2006.

2.4 | Data analysis

Mass, sample volumes, cell counts, and NeuN-positive percentages for each structure were entered into Excel spreadsheets for calculations. When more than one case was collected per species, average values (listed in Tables 1–3) were used and imported into JMP 14 Pro (SAS, USA) analysis software. For the purposes of this analysis, species were grouped into Pteropodidae (n = 7), Rhinolophoidae (n = 3), and Yangochiroptera (n = 3) according to the best current consensus (Hawkins et al., 2019). To test for differences in the calculated variables across groups, we used one-way ANOVA of the group then Wilcoxon nonparametric analysis of all pairs. Scaling relationships were calculated for the Pteropodidae sample using regressions to power functions; we then determined whether values for Rhinolophoidea and Yangochiroptera species fell within or outside the 95% confidence intervals for individual values, given the fitted functions. As in our previous studies, we chose not to use mathematical methods of "correcting" for phylogenetic relatedness in the dataset for three reasons. First, we already take the possibility of clade-specific features into account by analyzing our data separately by

TABLE 2 Numbers of neurons and neuronal density (neurons/mg)

Species	Micro/mega	Family	N_{CX}	N_{CB}	N_{RoB}	$D_{N,CX}$	$D_{N,Cb}$	$D_{N,RoB}$
Cardioderma cor	Microbat	Megadermatidae	10,221,440	63,792,242	7,317,510	39,766	616,497	36,367
Coleura afra	Microbat	Emballonuridae	5,080,428	47,984,992	5,514,794	56,460	888,050	55,816
Triaenops persicus	Microbat	Hipposideridae	6,039,272	59,651,487	4,072,165	64,765	820,968	47,981
Hipposideros commersoni	Microbat	Hipposideridae	7,877,834	51,875,702	6,965,297	32,406	483,525	29,267
Chaerephon pumilus	Microbat	Molossidae	5,632,618	25,324,375	4,001,397	66,967	697,515	47,232
Miniopterus schreibersii	Microbat	Miniopteridae	6,116,028	44,447,776	5,094,974	71,277	915,845	59,204
Epomophorus wahlbergi	Megabat	Pteropodidae	26,201,540	121,136,246	14,416,846	41,326	626,067	37,309
Rousettus aegyptiacus	Megabat	Pteropodidae	28,859,259	132,436,158	10,936,630	29,287	602,896	19,962
Megaloglossus woermanni	Megabat	Pteropodidae	11,897,169	49,734,021	4,097,177	38,490	627,122	25,755
Hypsignathus mostrosus	Megabat	Pteropodidae	24,343,593	221,298,619	28,877,041	13,481	484,806	24,684
Epomops franqueti	Megabat	Pteropodidae	22,533,501	150,660,138	12,808,860	21,663	525,829	17,986
Casinycteris argynnis	Megabat	Pteropodidae	10,471,765	77,445,413	11,098,900	22,666	605,042	36,751
Scotonycteris zenkeri	Megabat	Pteropodidae	12,847,119	47,217,961	5,529,891	39,652	549,046	31,067

TABLE 3 Fractional mass and numbers of neurons in cerebral cortex and cerebellum, shown as percentages of total mass or number of neurons in the brain

Species	Micro/mega	Family	M_{CX}/M_{BR}	N_{CX}/N_{BR}	M_{CB}/M_{BR}	N_{CB}/N_{BR}	M_{CX}/M_{CB}	N_{CB}/N_{CX}
Cardioderma cor	Microbat	Megadermatidae	45.7	12.6	18.3	78.4	2.50	6.24
Coleura afra	Microbat	Emballonuridae	36.9	8.7	22.5	81.9	1.64	9.45
Triaenops persicus	Microbat	Hipposideridae	37.5	8.7	29.1	85.5	1.29	9.88
Hipposideros commersoni	Microbat	Hipposideridae	41.0	11.8	18.2	77.8	2.25	6.59
Chaerephon pumilus	Microbat	Molossidae	40.4	16.1	17.3	72.4	2.33	4.50
Miniopterus schreibersii	Microbat	Miniopteridae	38.8	11.0	21.9	79.9	1.77	7.27
Epomophorus wahlbergi	Megabat	Pteropodidae	52.5	16.2	15.9	74.9	3.30	4.62
Rousettus aegyptiacus	Megabat	Pteropodidae	55.5	16.8	12.6	76.9	4.40	4.59
Megaloglossus woermanni	Megabat	Pteropodidae	56.2	18.1	14.7	75.7	3.81	4.18
Hypsignathus mostrosus	Megabat	Pteropodidae	52.4	8.9	13.3	80.6	3.94	9.09
Epomops franqueti	Megabat	Pteropodidae	50.8	12.1	14.0	81.0	3.63	6.69
Casinycteris argynnis	Megabat	Pteropodidae	51.8	10.6	14.3	78.2	3.61	7.40
Scotonycteris zenkeri	Megabat	Pteropodidae	55.1	19.6	14.6	72.0	3.77	3.68

clade; second, we do not want our results to depend on assumed phylogenetic relationships that may or may not withstand the test of new data; and third, the issues addressed here relate to whether particular variables related to distribution of mass and numbers of neurons across brain structures differ between Pteropodids and either Rhinolophoids or Yangochiroptera. We thus prefer to report all data untainted by assumed phylogenetic relationships that may or may not hold.

3 | RESULTS

The chiropterans in our sample had body masses ranging between 5.4 and 310.0 g and brains that varied between 0.208 and 3.459 g in mass (Table 1). These species comprise some of the smallest mammals and mammalian brains whose cellular composition we have analyzed to date. We find that our sample of chiropteran species have very small numbers of neurons in the cerebral cortex (between 5 and 30 million), cerebellum (25–221 million) and in the rest of brain (4–30 million; Table 2).

3.1 | Microchiropterans do not have more neurons than expected in the cerebellum

Nine of the 13 species analyzed here have similar brain mass in the range between 0.5 and 1.0 g: all three rhinolophoids, all three yangochiropterans, and three of the seven pteropodids, which makes them comparable, especially since the largest pteropodid brains were still small (Table 1). Still, and as expected (Maseko et al., 2012), the Yangochiroptera and Rhinolophoidea species in our sample show relatively large cerebella that amount to 20.6 ± 2.8% and 21.9 ± 6.2% of brain mass, respectively, which is significantly higher than the 14.2 ± 1.1% of brain mass occupied by the cerebellum in Pteropodidae (Wilcoxon, both $p = .0227$; one-way ANOVA, $p = .0085$; Table 3, Figure 1a). Conversely, the cerebral cortex occupies a smaller percentage of brain mass in both Yangochiroptera and Rhinolophoidea (38.7 ± 1.8% and 41.4 ± 4.2%, respectively) compared to Pteropodidae (53.5 ± 2.1%, again both $p = .0227$, Wilcoxon; one-way ANOVA, $p < .0001$; Table 3, Figure 1b). In turn, the rest of brain is only relatively smaller in Yangochiroptera (1.3 ± 1.5% of brain mass) compared to Pteropodidae (4.8 ± 2.1%, $p = .0227$), and has an intermediate relative size in Rhinolophoidae (3.3 ± 2.1% of brain mass) that is not significantly different from either of the other clades (both $p > .1$; one-way ANOVA, $p = .0019$; Figure 1c).

Strikingly, the relatively large cerebellum of Yangochiroptera and Rhinolophoidea compared to Pteropodidae does not hold a larger proportion of brain neurons (one-way ANOVA, $p = .4445$): in these clades, the cerebellum contains 78.1 ± 5.0%, 80.6 ± 4.3, and 77.0 ± 3.2% of all brain neurons (Table 3, Figure 1a, right). Percentages of brain neurons found in the cerebral cortex are also similar across Yangochiroptera, Rhinolophoidea, and Pteropodidae (11.9 ± 3.8%, 11.0 ± 2.1%, and 14.6 ± 4.1%, respectively; one-way ANOVA, $p = .3407$), as are the percentages of neurons in the rest of brain (10.0 ± 1.2%, 8.4 ± 2.4% and 8.4 ± 2.0%, respectively; one-way ANOVA, $p = .4762$). Thus, we find that the relatively larger cerebellum of Yangochiroptera and Rhinolophoidea microchiropterans does not have relatively more brain neurons than the cerebellum of Pteropodidae.

3.2 | Microchiropterans and megachiropterans share similar neuronal scaling rules within brain structures

Another way to state our finding regarding the distribution of brain neurons in chiropterans, quantified in Figure 1, is that for similar proportions of brain neurons as found in most other mammalian species,

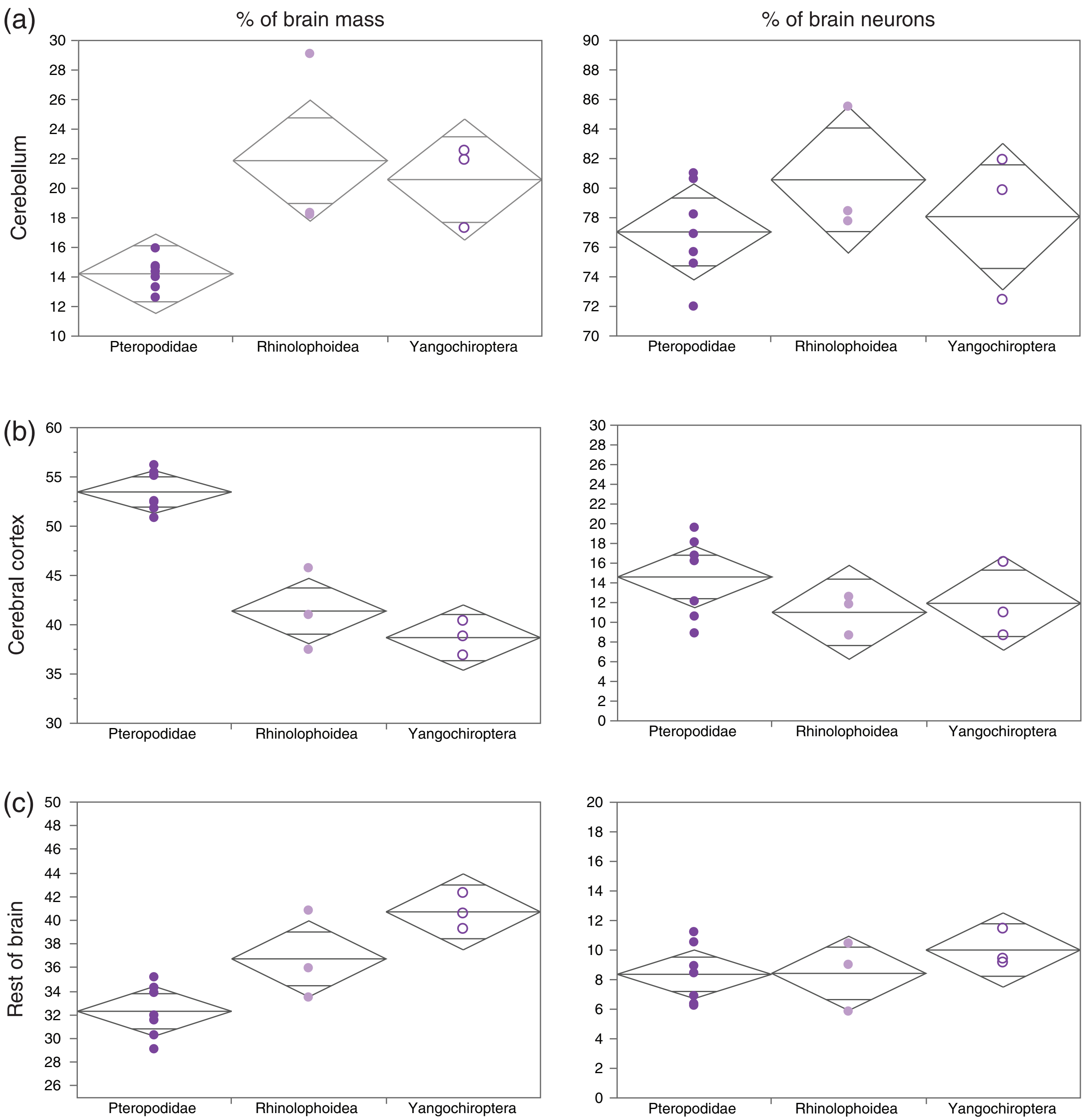

FIGURE 1 Small echolocating Rhinolophoidea and Yangochiroptera bats have relatively large cerebella and relatively small cortices compared to the Pteropodidae (megachiropterans), but similar proportions of brain neurons in each brain structure. Graphs show percent of total brain mass (left) and percent of total brain neurons (right) located in the cerebellum (a), cerebral cortex (b), and the rest of the brain (c) for each group: Pteropodidae (solid violet filled circles; *n* = 7), Rhinolophoidea (solid lavender circles; *n* = 3), and Yangochiroptera (unfilled violet circles; *n* = 3) samples. Diamond plots depict the 95% confidence intervals (CI) for each group, where the group mean is indicated by the central horizontal line, and the upper and lower bounds are represented by the top and bottom horizontal lines, respectively. Group averages are given in the main text [Color figure can be viewed at wileyonlinelibrary.com]

the cerebral cortex is relatively smaller (Figure 2a), and the cerebellum is relatively larger (Figure 2b), in Yangochiroptera and Rhinolophoidea compared to Pteropodidae and most other mammalian species (Figure 2). One possible explanation for this finding could be that the two former clades follow very particular brain scaling rules such that the relationships between structure mass and number of neurons that apply to their cerebral cortex and cerebellum are different from those that apply to other mammals, including Pteropodidae. To address this possibility, we calculated the scaling rules that apply to the cellular composition of the seven Pteropodidae species in our sample, and

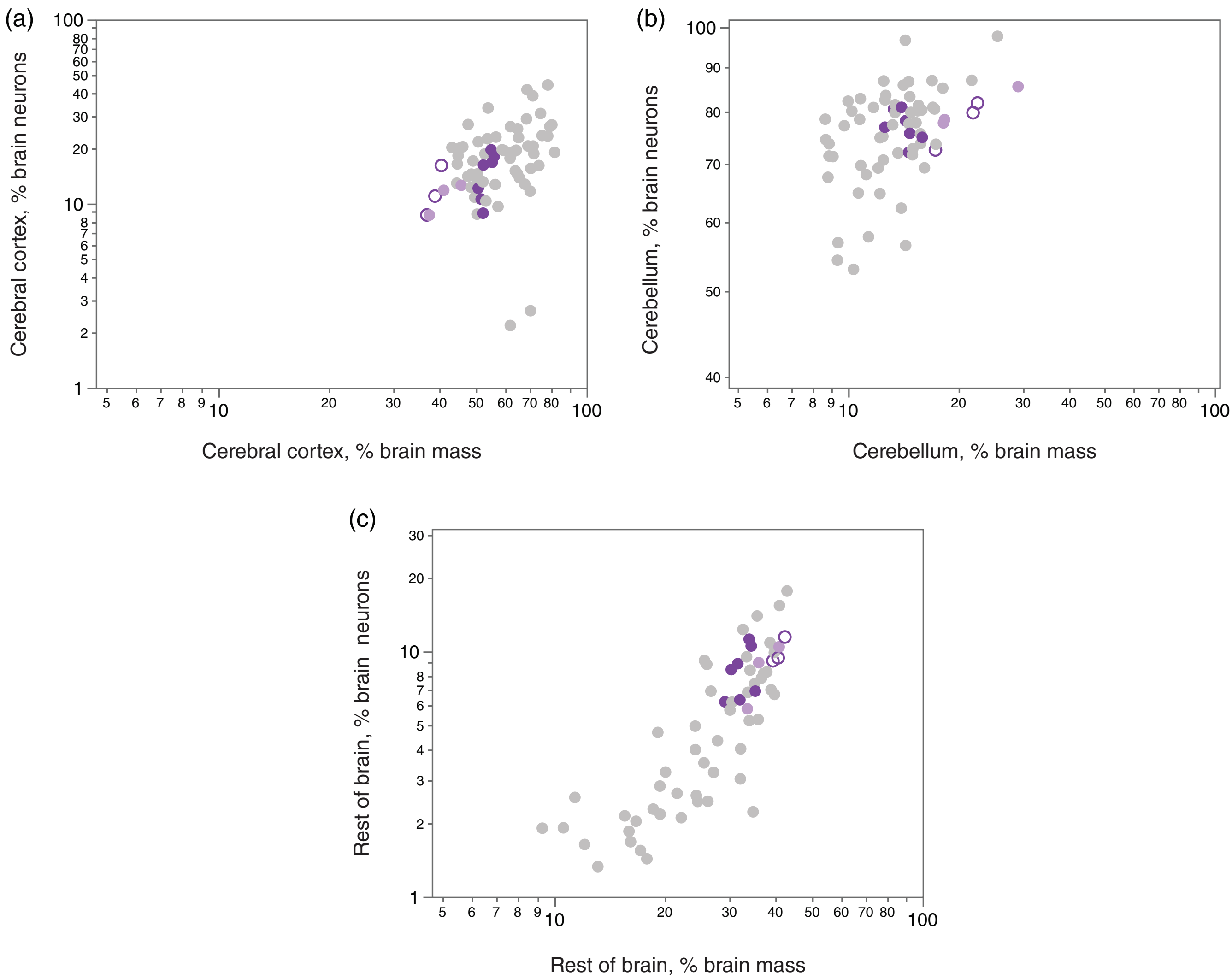

FIGURE 2 Rhinolophoidea and Yangochiroptera species have similar percentages of brain neurons in the cerebellum and cerebral cortex as the Pteropodidae and most other mammalian species. Graphs show how the percent of total brain neurons in each structure (cerebral cortex, a; cerebellum, b; rest of brain, c) vary with the percent of total brain mass occupied by the structure. Bat species are shown in shades of purple (Pteropodidae, solid purple circles; Rhinolophoidea, solid lavender circles; Yangochiroptera, unfilled violet circles). Gray circles depict values for other nonprimate mammalian species (marsupials, afrotherians, glires, scandentians, eulipotyphlans, carnivorans, and artiodactyls). We do not identify nonmammalian clades so as not to obscure the main comparison, which is across chiropteran species examined here. All x-axes are shown at the same scale for comparison across structures [Color figure can be viewed at wileyonlinelibrary.com]

compared to them the data points for species of the other two clades. Figure 3 shows that the scaling relationships that predict the mass of the major brain divisions in Pteropodidae species have exponents close to linearity and also predict brain structure mass in all Yangochiroptera and Rhinolophoidea species within the 95% confidence interval for individual values. Notice that the Pteropodidae neuronal scaling rules also predict structure mass by number of neurons relationships for most nonprimate mammalian species in the same range. Most importantly, we find that the proportionately larger cerebellum in the two test clades is not overly large for its number of neurons (Figure 3b); if anything, it is the mass of the cerebral cortex of all 3 Yangochiroptera and 2 of the 3 Rhinolophoidea that is somewhat smaller than expected for the number of cerebral cortical neurons in these species, although still within the 95% confidence interval for individual values (Figure 3a). The finding that the proportionately larger cerebellum of Yangochiroptera and Rhinolophoidea is neither an overly neuron-rich brain structure nor an enlarged cerebellum for its number of cerebellar neurons suggests an alternative possibility: that it is the cerebral cortex in these species that is relatively smaller than in Pteropodidae and other mammals, making the cerebellum appear overly large.

3.3 | Microchiropterans have a relatively smaller cerebral cortex, not an enlarged cerebellum

To test the possibility that the microchiropteran cerebellum appears enlarged because of a relatively small cerebral cortex, we

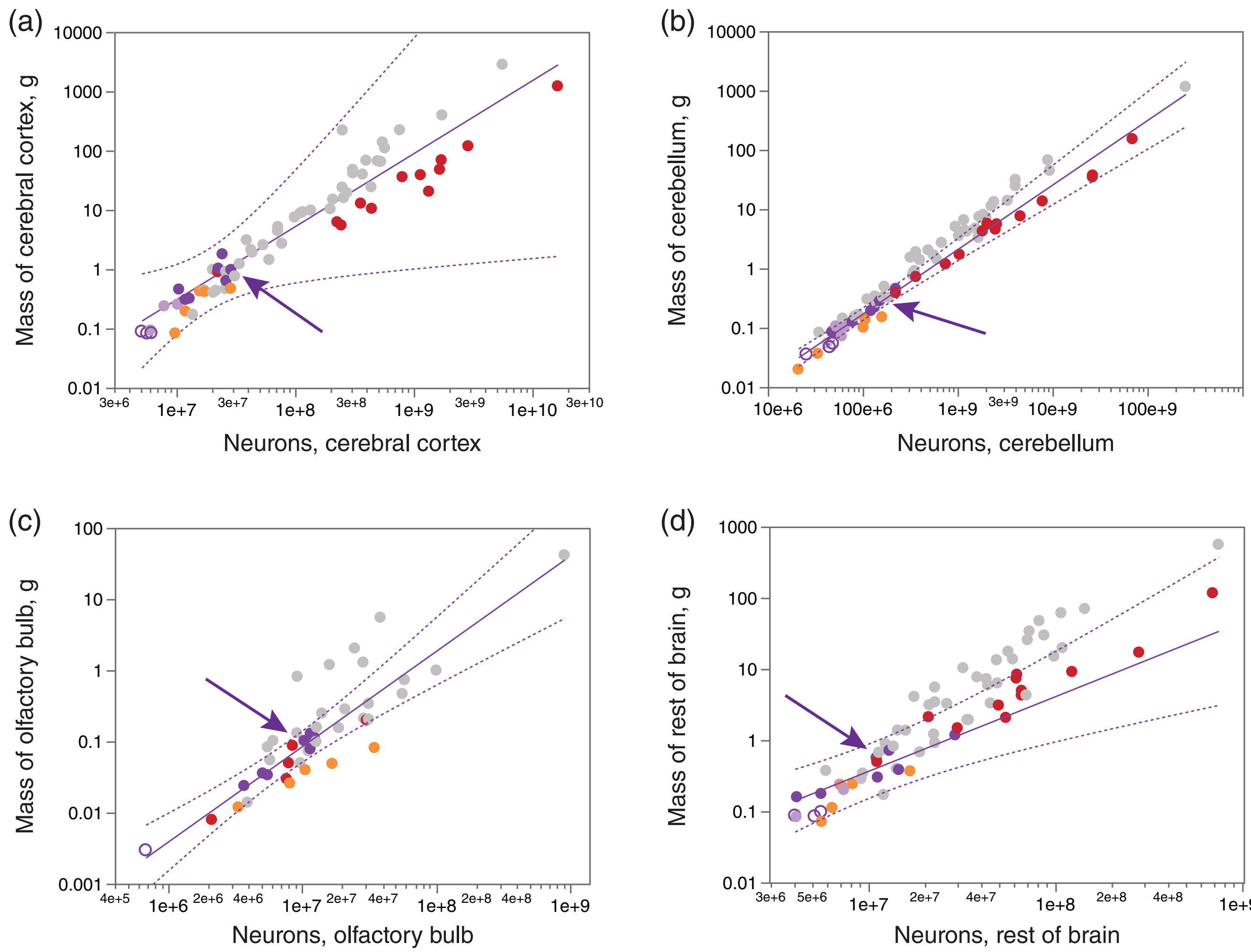

FIGURE 3 Rhinolophoidea and Yangochiroptera brain structures conform to the neuronal scaling rules that apply to Pteropodidae, which also overlap with other mammals. Each graph depicts the mass of each main brain structure (cerebral cortex, a; cerebellum, b; olfactory bulb, c; rest of brain, d) in each species and how it varies with the numbers of neurons in the same structure. Functions plotted refer to the seven Pteropodidae species in our present dataset (solid line) with dotted lines indicating the 95% confidence interval for individual values. Exponents for the plotted functions ± standard error are (a) 1.232 ± 0.420 (r^2 = .632, p = .0325), (b) 1.088 ± 0.061 (r^2 = .984, p < .0001), (c) 1.339 ± 0.149 (r^2 = .942, p = .0003), and (d) 1.049 ± 0.204 (r^2 = .841, p = .0036). Chiropteran species are shown in shades of purple (Pteropodidae, solid purple circles; Rhinolophoidea, solid lavender circles; Yangochiroptera, unfilled violet circles); eulipotyphlans are shown in orange; primate species are shown in red. Gray circles depict values for other mammalian species (marsupials, afrotherians, glires, scandentians, carnivorans, and artiodactyls). In each graph, the purple arrow indicates *Rousettus aegyptiacus*, the only echolocating Pteropodidae in our sample. Notice that primate data points (red) fall well within the 95% CI for individual values calculated for Pteropodidae [Color figure can be viewed at wileyonlinelibrary.com]

examined the scaling of mass and numbers of neurons directly across brain structures, again using the Pteropodidae as reference. We find that while the scaling relationship between cerebellar and cerebral cortical masses in Pteropodidae overlaps very well with other mammalian species, all six Yangochiroptera and Rhinolophoidea sit outside the 95% confidence interval for the scaling relationship (Figure 4a). Again, these could be described as overly large cerebella for the mass of the corresponding cerebral cortex, or as too small cerebral cortices for the mass of the corresponding cerebellum. Using the rest of the brain as reference, we find that it is the cerebral cortex of Yangochiroptera and Rhinolophoidea species that is significantly smaller than predicted for the mass of the rest of the brain in the same species (Figure 4b), while the cerebellum has the predicted mass for the mass of the rest of brain (Figure 4c).

Remarkably, and in agreement with the similar proportions of brain neurons in each structure across the three clades, we find that the proportionality between numbers of neurons in the cerebellum and in the cerebral cortex that applies to Pteropodidae also predicts numbers of neurons in both structures in Yangochiroptera and Rhinolophoidea alike (Figure 4d), and so do the scaling relationships between numbers of neurons in the rest of the brain and either the cerebral cortex (Figure 4e) and cerebellum (Figure 4f). These data thus indicate that the relatively large cerebellum of Yangochiroptera and Rhinolophoidea bats is not an enlarged cerebellum, or a cerebellum with a larger absolute or relative number of neurons, but rather the result of a comparatively diminutive cerebral cortex. Incidentally, these relationships also show that while the neuronal scaling rules within each brain structure in Pteropodidae might extend to primates (Figure 3), the Pteropodidae do not share with primates the neuronal scaling rules that apply across primate brain structures (Figure 4).

We further analyze this possibility by doing crossed comparisons between structures. Again, we confirm that the mass of the cerebral cortex in Pteropodidae is well predicted by numbers of neurons in the cerebellum (Figure 5a) and in the rest of the brain (Figure 5b) of these animals, in a relationship that also includes most other mammalian species in the same range (solid gray circles) but excludes most of the Yangochiroptera and Rhinolophoidea bats (unfilled violet and solid lavender circles), whose cerebral cortices are smaller than predicted for their numbers of neurons in the other brain structures. The converse analysis, in turn, shows that the relationships that predict the mass of the cerebellum from the numbers of neurons found in the cerebral cortex and rest of brain of Pteropodidae also predict well the

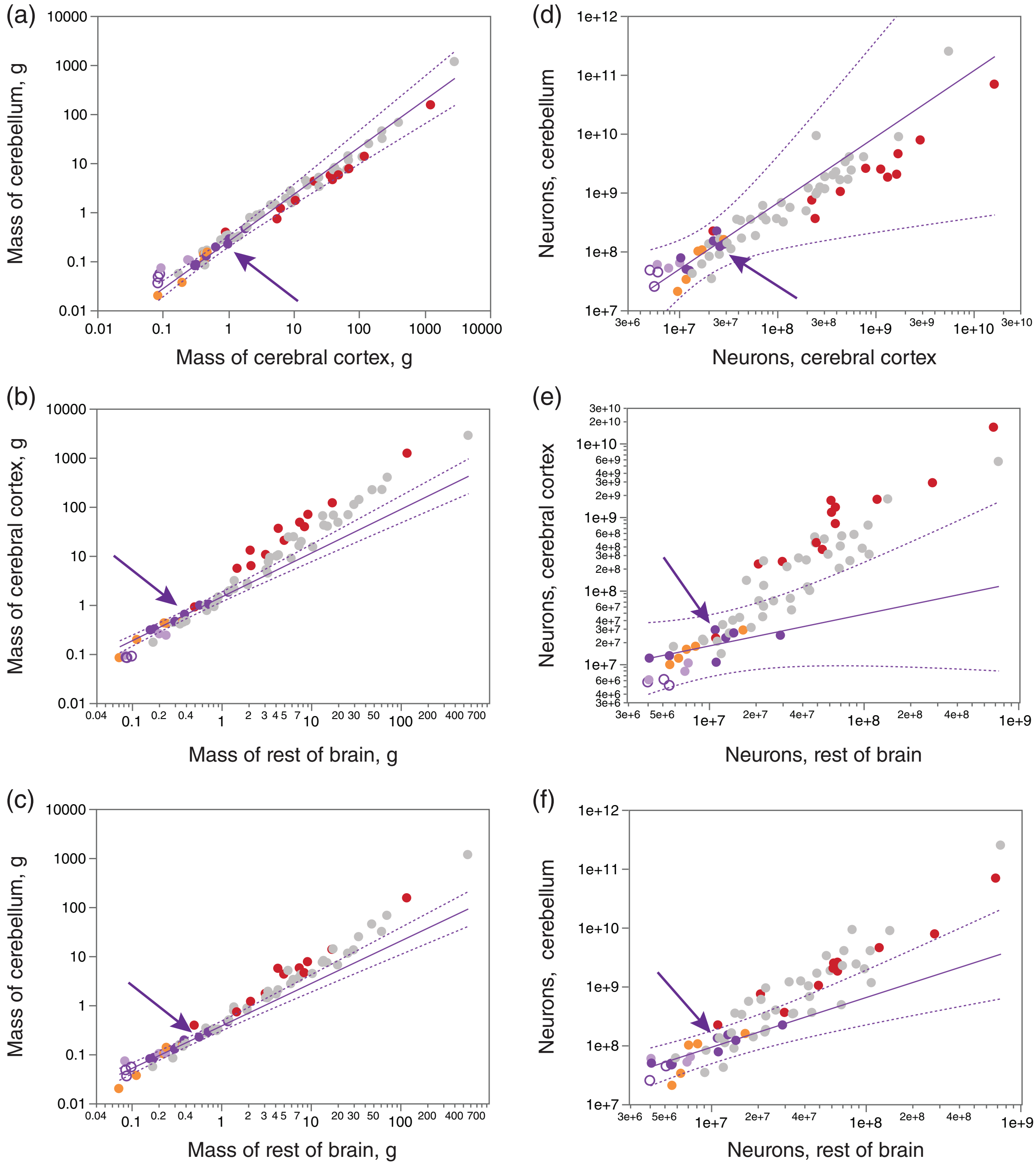

FIGURE 4 Legend on next page.

mass of the cerebellum in Yangochiroptera and Rhinolophoidea species (Figure 5c,d).

3.4 | Microchiropterans small body sizes predict diminutive cerebral cortex

Finally, we examined whether the diminutive body mass of Yangochiroptera would be a simpler explanation for their reduced cerebral cortical mass compared to Pteropodidae. We again use the expedient of calculating the scaling rules that apply to Pteropodidae, and comparing the other chiropteran species in our sample to them, as well as the eulipotyphlans in our dataset (Figure 6, shown in orange), which overlap in range of body mass with the chiropterans in our sample (below 10 g to over 100 g). We find that all four bat species with a significantly smaller cerebral cortex compared to the prediction for Pteropodidae (i.e., all three Yangochiroptera plus the Rhinolophoid *Triaenops persicus*) share the distinction of being the smallest animals in our entire mammalian dataset, together with the smallest Eulipotyphlan (*Sorex fumeus*), all below 15 g body mass (Figure 6a). Additionally, all four smallest bat species as well as the diminutive smoky shrew have distinctively the smallest cerebral cortices among all animals in our extended dataset (Figure 6a). In contrast, by the same metric of comparing to body mass, echolocating microchiropterans do not have an enlarged cerebellum (Figure 6b), and the mass of the rest of the brain also scales with body size as it does in Pteropodidae (Figure 6c). Importantly, there is no evidence of significantly decreased numbers of neurons in any brain structure in these tiniest of mammals relative to the predicted numbers for body mass (Figure 6d–f). Our results thus indicate that the diminutive cerebral cortex of microchiropterans, as in the tiny smoky shrew, may be a feature associated with very small bodies, or may suggest an additional morphological character indicative of phylogenetic affinities between the microchiropterans and shrews (Dell et al., 2010).

4 | DISCUSSION

The current study reveals findings of specific interest to the questions raised regarding the seemingly enlarged cerebellum in microchiropterans (Yangochiroptera and Rhinolophoidea, in our current sample) and the phylogenetic affinities of the microchiropterans and megachiropterans to each other and to other mammalian species. We find that the disproportionately large cerebellum of Yangochiroptera and Rhinolophoidea does not consist of a disproportionately large number of neurons; rather, it has the same 75–80% of all brain neurons found in pteropodids and other mammals. Further, we find that all three bat clades analyzed share with each other, as well as with primates, the scaling relationship between brain structure mass and number of neurons in the structure. All three bat clades analyzed also share scaling relationships across brain structures, but here they stand apart from primates. Most importantly, using the rest of brain as a reference, we find that the mass of the cerebral cortex of the yangochiropteran and rhinolophoid species is smaller than would be expected, while the mass of the cerebellum is not enlarged. We would like to point out at this point that clade-specific allometric effects due to brain size variation are highly unlikely in the present study. Unlike our previous studies that included animals of a wide range of brain masses, all six species classified separately here as yangochiropteran or rhinolophoids plus three pteropodids have similar brain masses of between 0.2 and 1.0 g. The four remaining species still had brains concentrated in the 1.0–3.4 g range. Thus, the overall variation in brain mass barely encompasses one order of magnitude. We are thus comparing like to like, which we feel that also justifies the simple nonparametric analysis employed here so as not to assume normality in the

FIGURE 4 Yangochiroptera and Rhinolophoidea species have a smaller than expected cerebral cortical mass for their cerebellar mass and rest of brain mass, but a regular sized cerebellum compared to the rest of brain. Scaling of mass (a–c) and numbers of neurons (d–f) across brain structures: Cerebellum versus cerebral cortex (a,d), cerebral cortex versus rest of brain (b,e) and cerebellum versus rest of brain (c,f). The power function plotted in purple (solid line; confidence interval depicted by dotted lines) was calculated for Pteropodidae species (n = 7, solid purple circles) as a reference against which to compare other chiropteran species (Rhinolophoidea, solid lavender circles; Yangochiroptera, unfilled violet circles) and mammals (primates, red; eulipotyphlans, orange; marsupials, afrotherians, glires, scandentia, carnivorans, and artiodactyls in gray). (a) Cerebellar mass scales across Pteropodidae species with cortical mass raised to an exponent of 0.961 ± 0.058 (r^2 = .982, p < .0001), and this scaling excludes all three Yangochiroptera and all three Rhinolophoidea. (b) Mass of cerebral cortex scales across Pteropodidae species with rest of brain mass raised to an exponent of 0.893 ± 0.042 (r^2 = .989, p < .0001), and all three Yangochiroptera and three Rhinolophoidea species have significantly smaller cerebral cortices than expected for a Pteropodidae with their rest of brain mass. (c) Scaling of Pteropodidae cerebellar mass with rest of brain mass raised to an exponent of 0.866 ± 0.042 (r^2 = .988, p < .0001) also predicts cerebellar mass of all three Yangochiroptera and two of three Rhinolophoidea; only *Triaenops persicus* has a larger than predicted cerebellar mass. (d–f) In contrast to scaling of mass, the scaling of numbers of neurons across brain structures in Pteropodidae species is also predictive of numbers in all Yangochiroptera and Rhinolophoidea species. (d) numbers of cerebellar neurons scale with numbers of neurons in the cerebral cortex as a power function of exponent 1.124 ± 0.350 (r^2 = .674, p = .0236), not significantly different from linearity. (e) Yangochiroptera and Rhinolophoidea appear to have fewer neurons in the cerebral cortex than expected for the numbers of neurons in their rest of brain, but the scaling of numbers of cerebral cortical neurons with numbers of neurons in the rest of brain does not reach significance across Pteropodidae (exponent 0.432 ± 0.225, r^2 = .424, p = .1132). (f) numbers of cerebellar neurons scale with numbers of neurons in the rest of brain as a power function of exponent 0.845 ± 0.149 (r^2 = .866, p = .0024), not significantly different from linearity. In each graph, the purple arrow indicates *Rousettus aegyptiacus*, the only echolocating Pteropodidae in our sample. Note that here, Pteropodidae no longer share scaling rules with primates (red) [Color figure can be viewed at wileyonlinelibrary.com]

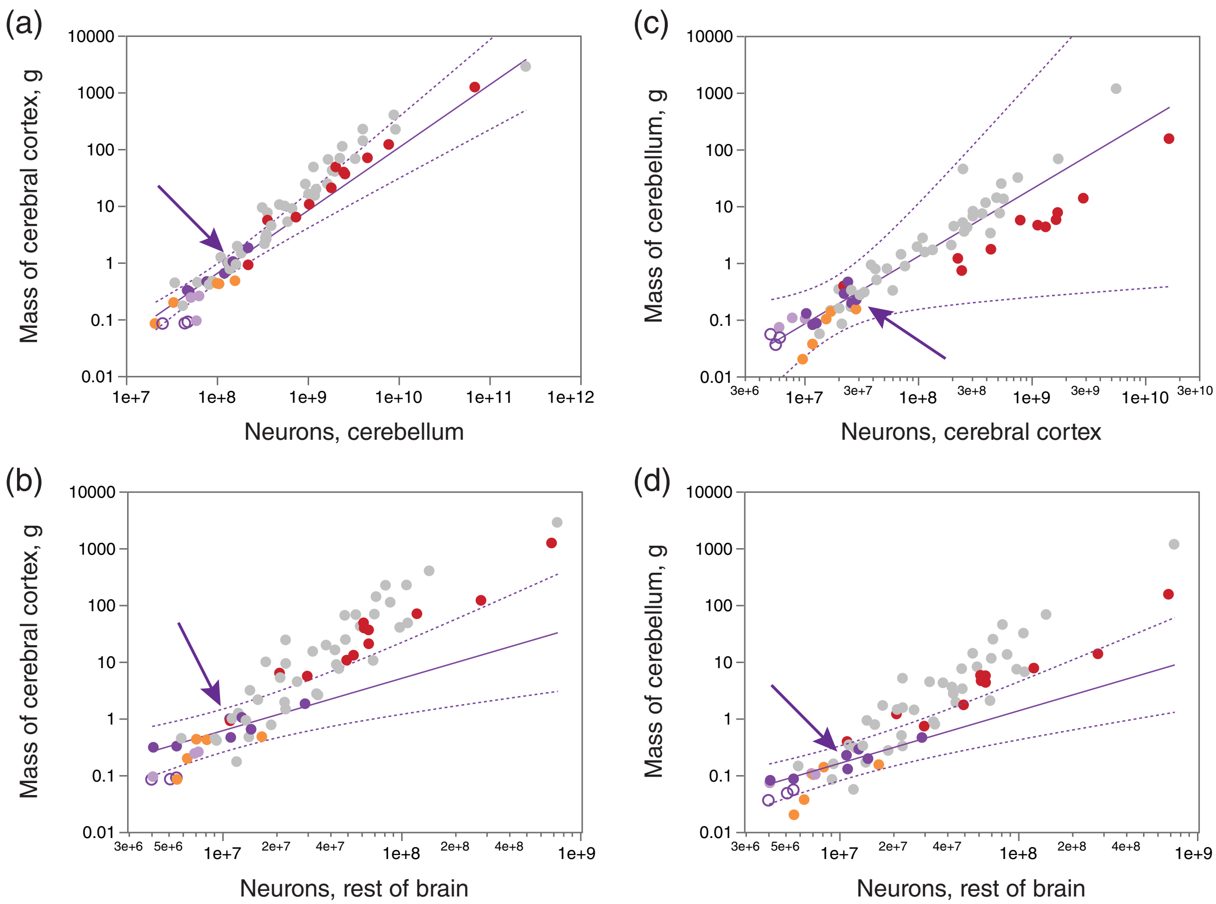

FIGURE 5 Echolocating microchiropterans have a diminished cerebral cortex, not an enlarged cerebellum: Across all three Yangochiroptera and in one of three Rhinolophoidea (*Triaenops persicus*), the mass of the cerebral cortex is significantly smaller than predicted from Pteropodidae, while cerebellar mass is not significantly larger than predicted. Crossed comparisons of scaling between masses of test brain structures (cerebral cortex, a,b; cerebellum, c,d) as functions of numbers of neurons in reference brain structures (cerebellum, a; cerebral cortex, c; rest of brain, b,d). Scaling functions calculated for Pteropodidae species (n = 7; solid line; dotted lines, 95% confidence interval). (a) Cerebral cortical mass scales across Pteropodidae species with number of cerebellar neurons raised to an exponent of 1.109 ± 0.100 (r^2 = .961, p < .0001), not significantly different from linearity. Yangochiroptera and one Rhinolophoidea species (*Triaenops persicus*) have significant smaller cerebral cortices than expected for their numbers of neurons in the cerebellum. (b) As in a, cerebral cortical mass scales across Pteropodidae species with number of rest of brain neurons raised to an exponent of 0.921 ± 0.202 (r^2 = .805, p = .0061), also not significantly different from linearity, and all three Yangochiroptera and one Rhinolophoidea species (*Triaenops persicus*) have significant smaller cerebral cortices than expected for their numbers of neurons in the rest of brain. c,d: In contrast, the scaling functions calculated for Pteropodidae for mass of the cerebellum with number of neurons in the cerebral cortex (c, exponent 1.188 ± 0.411; r^2 = .626, p = .0341) and in the rest of brain (d, exponent 0.926 ± 0.164, r^2 = .865, p = .0024) also predict cerebellar mass in all Yangochiroptera and Rhinolophoidea species tested. Solid violet circles, Pteropodidae; solid lavender circles, Rhinolophoidea; unfilled violet circles, Yangochiroptera; orange, eulipotyphlans; red, primates; filled gray circles, other mammalian species in our dataset. In each graph, the purple arrow indicates *Rousettus aegyptiacus*, the only echolocating Pteropodidae bat in our sample [Color figure can be viewed at wileyonlinelibrary.com]

distribution. In doing so, we find that yangochiropteran and rhinolophoid thus differ from pteropodid bats and the vast majority of other mammals previously studied in having a cerebral cortex of very small absolute size, like the smoky shrew, the only other animal in our dataset with a body mass less than 15 g. These findings are discussed in terms of the proposed enlargement of the microchiropteran cerebellum in relation to echolocation and the phylogenetic affinities of the microchiropterans and megachiropterans to each other and other mammalian species.

Here we show that the large relative size of the cerebellum in Yangochiroptera and Rhinolophoidea (all of them small echolocating bats once classified as Microchiroptera), amounting to ca., 21% of brain mass in contrast to the 14% found in Pteropodidae (once Megachiroptera) and in most other mammals, cannot be considered an evolutionarily enlarged cerebellum, but rather the result of the diminutive mass of the Yangochiroptera and Rhinolophoidea cerebral cortex compared to other mammals. Thus, although it remains strictly correct to state that the relative size of the cerebellum is larger in Yangochiroptera and Rhinolophoidea compared to Pteropodidae bats and other mammals, this statement masks the more accurate description, whose functional implications must now be sought: that the Yangochiroptera and Rhinolophoidea analyzed here, and possibly

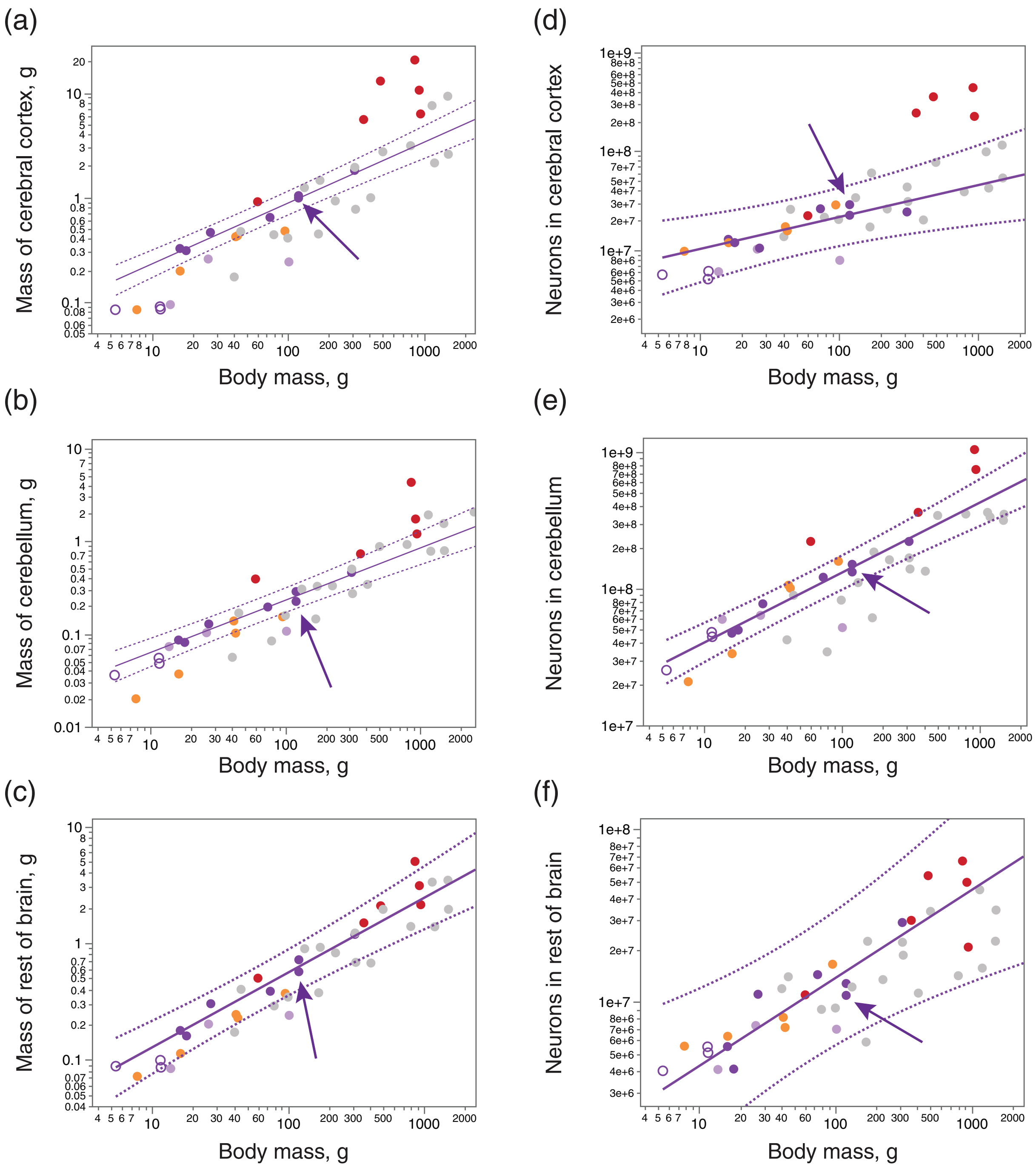

FIGURE 6 All Yangochiroptera as well as the smallest Rhinolophoidea (*Triaenops persicus*) and the smallest Eulipotyphlan species in our dataset (*Sorex fumeus*) have a diminished cerebral cortex, not an enlarged cerebellum, compared to Pteropodidae, but show no evidence of significantly decreased numbers of neurons in any brain structure. Graphs show scaling of mass (a–c) or numbers of neurons (d–f) in brain structures (cerebral cortex, a,d; cerebellum, b,e; rest of brain, c,f) as power functions of body mass calculated for Pteropodidae species (n = 7; solid line; dotted lines, 95% confidence interval). Exponents: a, cerebral cortical mass versus body mass, 0.582 ± 0.034 (r^2 = .983, p < .0001); b, cerebellar mass versus body mass, 0.563 ± 0.040 (r^2 = .976, p < .0001); c, rest of brain mass versus body mass, 0.640 ± 0.059 (r^2 = 0.959, p = .0001); d, cerebral cortical neurons versus body mass, 0.323 ± 0.088 (r^2 = .727, p = .0148); e, cerebellar neurons versus body mass, 0.512 ± 0.038 (r^2 = .974, p < .0001); f, rest of brain neurons versus body mass, 0.509 ± 0.117 (r^2 = .791, p = .0074). Solid violet circles, Pteropodidae; solid lavender circles, Rhinolophoidea; unfilled violet circles, Yangochiroptera; filled orange circles, Eulipotyphla (Herculano-Houzel, Kaas, et al., 2015; Sarko et al., 2009); red, primates (Azevedo et al., 2009; Gabi et al., 2010; Herculano-Houzel et al., 2007; Herculano-Houzel, Kaas, et al., 2015); gray circles, other mammalian species in our dataset (Dos Santos et al., 2017; Herculano-Houzel, Kaas, et al., 2015; Jardim-Messeder et al., 2017). In each graph, the purple arrow indicates *Rousettus aegyptiacus*, the only echolocating Pteropodidae in our sample [Color figure can be viewed at wileyonlinelibrary.com]

small echolocating bats as a whole, exhibit a minuscule cerebral cortex. We show that this is the case first by establishing that the disproportionately large cerebellum of Yangochiroptera does not hold a disproportionately large percentage of brain neurons; instead, we find that Yangochiroptera, Rhinolophoidea and Pteropodidae share with each other and many other mammals the same linear relationship between numbers of neurons in the cerebellum and cerebral cortex. Next, by using the remaining areas as an external reference to which the cerebral cortex and cerebellum can both be compared, we reveal that the cerebellum of Yangochiroptera and of the smallest Rhinolophoid in our dataset is only as large in mass as expected for Pteropodidae, while the cerebral cortex is smaller than expected. Third, we show that all brain structures in the small echolocating bats analyzed have the mass that would be expected for Pteropodidae (and other mammals) with their numbers of neurons, which indicates that the scaling rules that apply within each brain structure remain fairly similar across the two clades, although all Yangochiroptera as well as the smallest Rhinolophid tend to have smaller cerebral cortices than expected for their numbers of cortical neurons. Fourth, we show that the cerebellum of the Yangochiroptera and Rhinolophoid bats in our sample is only as large as expected for Pteropodidae and other mammals given the numbers of neurons in the cerebral cortex and rest of brain of each species, while the cerebral cortex of Yangochiroptera as well as of the smallest Rhinolophoid is smaller than expected for numbers of neurons in cerebellum and rest of brain. Finally, we show that all animals in our dataset with body mass below 15 g, which include all three Yangochiroptera, one Rhinolophoidea (*Triaenops persicus*) and the smallest Eulipotyphla (*Sorex fumeus*), have diminutive cerebral cortices of less than 0.1 g, well below the predicted mass for the scaling relationship that applies to Pteropodidae as well as other mammals (all larger than 15 g).

Much has been speculated about the reason and role of the relatively large cerebellum of microbats, solely from relative size (e.g., Maseko et al., 2012). Our data show directly that Yangochiroptera, all of which were once classified as microbats, have neither enlarged cerebella with larger than usual fractions of brain neurons nor absolutely larger than expected numbers of cerebellar neurons, discrediting the functional significance of their relatively large cerebellum in relation to echolocation. Similarly, here we find that the brain of *Rousettus aegyptiacus*, the only echolocating Pteropodidae species in the dataset, does not stand out from others in its clade in the mass or number of neurons of any of the brain structures examined. While the midbrain colliculi, considered to be distinctively large in small echolocating bats (Jepsen, 1966), was not analyzed separately in the present study, it was quantified as part of the "rest of brain," which we found not to be systematically too large nor too small compared to the relationship that applies to Pteropodids, nor to have extraordinarily larger numbers of neurons.

Instead of speculating on their relatively large cerebellum, then, one must now consider why the *cerebral cortex* of Yangochiroptera is *smaller* in three-dimensional size than expected for Pteropodidae. Arguments for functional specializations in evolution are usually made for relatively enlarged, not reduced, structures. One might still insist that the diminutive cerebral cortex is related to echolocation in Yangochiroptera, but that argument does not make functional sense given the known relative magnification, not shrinkage, of the auditory cortex in these echolocating animals (Suga & Jen, 1976). Similarly, as pointed out above, the echolocating Pteropodidae *Rousettus aegyptiacus* shows no evidence of a reduced cerebral cortex (or of an enlarged cerebellum, for that matter). Rather, the evidence for a smaller than expected cerebral cortex in the smallest Yangochiroptera and Rhinolophoidea (*Triaenops persicus*), as well as the smallest eulipotyphlan (*Sorex fumeus*), comprising all of the animals below 15 g body mass, makes it more parsimonious to suspect that echolocation is not related to evolutionary changes in the relative distribution of neurons in their brains, and also not to the diminutive mass of the cerebral cortex. Instead, the cerebral cortex might be relatively small in mass in these species (compared to other mammals) simply due to their diminutive body size.

Although evolution still is often equated with expansion of the cerebral cortex, this structure already represents over 50% of brain mass in the Pteropodidae analyzed here. Our finding that it is not the case that the brain as a whole is "miniaturized" in the smallest animals, but rather only the cerebral cortex that is smaller in mass (not numbers of neurons) than expected for other bats, suggests that something about the cerebral cortex in particular makes it particularly sensitive or vulnerable to the costs associated with a tiny body mass. It may appear paradoxical to consider that the smallest animals might be energetically constrained by their diminutive body mass, when we have proposed in the past that the large bodies of great apes might preclude them from having larger numbers of cortical neurons (Fonseca-Azevedo & Herculano-Houzel, 2012). However, as we pointed out in that study, the balance between, on the one hand, increased energy intake per unit time with larger body mass, and on the other, the increased energetic cost of larger body mass, results in an energetic trade-off when body mass becomes very large (as in great apes), but in a trade-in when body mass is tiny (as seen here), probably due to the inefficiency of ingesting calories while awake that is consequence of having a diminutive mouth. A compounding problem is that the smallest mammals tend to sleep most hours of the day, leaving very little waking time to feed (Herculano-Houzel, 2015).

Importantly, the true evolutionary scenario might be even simpler than that: if the earliest bats were diminutive creatures with diminutive cortices, as in the oldest known complete fossil, *Icaronycteris index* (Jepsen, 1966), then modern small echolocating bats might have simply retained those features. This is in contrast to an earlier conclusion based on mathematical modeling of evolution that "the ancestor of modern bats was intermediate in body, wing and brain sizes" (Safi, Seid, & Dechmann, 2005), which did not take the fossil record into consideration. In the case that ancestral bats were diminutive, which is consistent with recent propositions for early mammalian evolution as a whole (Rowe, Macrini, & Luo, 2011), the larger cerebral cortex of Pteropodidae as well as other extant mammals would be the derived feature, possibly affordable once body size became large enough that enough calories could be ingested (Herculano-Houzel, 2015). This scenario is consistent with the good overlap between the smallest bat species examined here and the smallest eulipotyphlan in our dataset, *Sorex fumeus*, in their relationships between body mass, mass of brain structures, and numbers of neurons in those structures. Since *Sorex* are not known to echolocate, it does not seem that echolocating adds to the energetic cost of the tiniest bats.

The preceding discussion addresses the issue of the small cerebral cortex in the small echolocating bats from the monophyletic view of chiropteran affinities, but a different scenario emerges if one applies the diphyletic hypothesis. Briefly, the diphyletic hypothesis of chiropteran affinities posits that the megachiropterans are a branch of the order Dermoptera, the closest extant relatives of the primates (Pettigrew, 1996; Pettigrew et al., 1989), while the microchiropterans are a branch of the family Soricidae (the true shrews, which form the most speciose radiation of the Eulipotyphla; Dell et al., 2010). Our results, specifically the identification of the small cerebral cortex in Yangochiroptera and Rhinolophoid species, argue in favor of a morphological split between micro- and megachiropterans, but more data on other microchiropterans currently classified as Yangopteropodidae will be necessary to establish whether their brain morphology is similar to that of Pteropodidae or not. Unfortunately, similar data on the Dermopteran brain are lacking, which does not allow us to compare megachiropterans to them. While our data do align the microchiropterans (Yangochiroptera and Rhinolophoid species) with the smallest shrew in our dataset, which could be construed as supporting the alignment of the microchiropterans with the Soricidae, it remains possible that these two groups share not phylogeny, but simply the potential convergent effect of small body size on the size of the cerebral cortex as discussed above. Obtaining similar data to that presented here across a broader range of species of potential phylogenetic interest in an overlapping range of body sizes, including Dermoptera, Scandentia, Eulipotyphla and a much wider range of Chiroptera species will provide a clearer picture of both the phylogenetic affinities of the Chiroptera, as well as of the evolution of echolocation in Chiroptera, besides improving our understanding of the relationship between brain and body size in mammalian evolution.

We are aware that many researchers in the field use correction for phylogenetic relatedness when reporting scaling relationships and may want to cast doubt on the present findings for our refusal to use the mathematical expedients that they consider mandatory. We disagree that we are not taking phylogenetic relationships into consideration: we do that when we use the current consensus of classification into Yangochiroptera, Rhinolophoidea and Pteropodidae to analyze these species in our dataset separately. What we choose to not do is adjust any reported exponents according to the presumed phylogenetic relatedness *within* each clade analyzed. We suspect that the essence of our disagreement with "correcting" for phylogenetic relatedness is that while users of these methods implicitly expect scaling relationships to reflect idiosyncrasies in the composition of each species' brain that depend on their evolutionary history and therefore relatedness, our approach is to first seek to identify scaling relationships that might apply universally across the closely related species that compose a clade, whatever the precise relationship among them, but differ across clades. In contrast to phylogenetic correction methods, we report the scaling exponents that truly describe the relationship between numbers of neurons that compose a structure and the size of that structure, as obtained from the available raw data, given their clustering in clades. This way we maintain our data analyses as free of assumptions as possible. In particular, our expedient of comparing each of the six "microchiropteran" species against the raw scaling relationship found to apply to the unchallenged monophyletic clade of pteropodids ascertains that our findings remain correct and applicable whatever the exact phylogenetic relationships across these species turn out to be, and also whether differences in scaling turn out to be due to grade shifts and/or allometric shifts. Importantly, because we make our data available in Tables 1–3, any researcher who is interested in examining the effect of particular alternative phylogenetic relationships on the data is free to do so—and to redo those analyses any time that the current consensus on those relationships changes. To be clear, we do not claim that this is the only correct way of doing the analysis; ours is an analysis that focuses on elucidating fundamental scaling relationships first, so that deviations that depend on evolutionary history and therefore phylogenetic relatedness can be better understood as a complementary, further step.

Finally, and especially in the context of the anticipated criticisms regarding the absence of correction for phylogenetic relatedness in our dataset, we note that, except for what might be effects of small body size, our analysis does not show any clear split across the species investigated that would support the recent classification of chiropterans into Yangochiroptera and Yinpterochiroptera; so far, our results still match the older classification of bats into microchiropterans and megachiropterans, which in turn is potentially consistent with both the monophyly and diphyly arguments of chiropteran evolution. Importantly, we find that chiropterans share with primates the neuronal scaling rules that apply to their brain structures, as would be expected if these two groups were evolutionarily related—although in that case, chiropterans and primates would have diverged in the distribution of mass and neurons across brain structures. We are currently investigating how taking quantitative brain anatomy into consideration impacts our understanding of the evolution and cladistics of bats.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

AUTHOR CONTRIBUTIONS

Suzana Herculano-Houzel and Paul R. Manger conceived the study; Paul R. Manger, Consolate Kaswera-Kyamakya, and Emmanuel

Gillissen collected animals and prepared the brains; Suzana Herculano-Houzel and Felipe Barros da Cunha processed tissue; Suzana Herculano-Houzel, Felipe Barros da Cunha, and Jamie L. Reed analyzed data; Felipe Barros da Cunha wrote the initial version of the manuscript; Jamie L. Reed contributed to the final version of the manuscript; Suzana Herculano-Houzel and Paul R. Manger wrote the final version of the manuscript.

PEER REVIEW

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available in Tables 1–3, and from the corresponding author in xls format upon request.

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